ISyE 6669 HW 14

Fall 2025

Note: There may be more than one correct way to model constraints in this homework. Verify that the formulation models the logical relations correctly and are linear (with integer variables).

1. Workers at the Execo manufacturing plant work 5 consecutive days a week, e.g. a worker can start work on a Monday and continue till Friday while another can start work on a Wednesday and continue till Sunday etc. Workers who are off on weekends (Saturday and Sunday) are paid \$300 per day (for each of their 5 work days), workers who's schedules involve one weekend day are paid \$320 per day (for each of their 5 work days), and workers who's schedules involve both weekend days are paid \$330 per day (for each of their 5 work days). The goal is to determine the number of workers for each schedule to minimize cost subject to the following restrictions:

   - There must be at least 25 workers each day.
   - There can be at most 35 workers on at least 3 of the 7 days of the week.
   - If the number of workers that start on a Monday exceed 10 then the number of workers available on a Wednesday should be no more than 28.
   - The number of workers that start on a Saturday should be less than the number of workers that start on a Monday or a Tuesday.
   - A total 40 workers are available.

   Formulate the above problem as a mixed-integer linear problem. Use x_i to denote the number (integer) of workers who start work on day i. You may need to introduce additional binary variables. If you use big-M numbers, please provide your best estimates of their values.

   Solution:

   Let's introduce following variables:

   - $x_i \in \mathbb{Z}_+,\ i = 1, \ldots, 7$ denote the number of workers who start working on day i, (1 is for Monday, 2 is for Tuesday, and so forth).

- $y_i \in \{0,1\},\ i = 1,\ldots,7$ indicates whether the number of workers exceeds 35 on day i (1 for yes and 0 for no).
- $z \in \{0,1\}$ indicates whether the number of workers that start on a Monday exceeds 10 (1 if yes, 0 if no)
- $w \in \{0,1\}$ indicates that the number of workers that start on a Monday is greater than the number of workers that start on a Tuesday (0 if yes, 1 if no).

Then, we can model given constraints as:

- There must be at least 25 workers each day:

$$
\begin{aligned}
x_1 + x_2 + x_3 + x_4 + x_5 &\geq 25 \\
x_2 + x_3 + x_4 + x_5 + x_6 &\geq 25 \\
x_3 + x_4 + x_5 + x_6 + x_7 &\geq 25 \\
x_4 + x_5 + x_6 + x_7 + x_1 &\geq 25 \\
x_5 + x_6 + x_7 + x_1 + x_2 &\geq 25 \\
x_6 + x_7 + x_1 + x_2 + x_3 &\geq 25 \\
x_7 + x_1 + x_2 + x_3 + x_4 &\geq 25
\end{aligned}
$$

- There can be at most 35 workers on at least 3 of the 7 days of the week:

$$
\begin{aligned}
x_1 + x_2 + x_3 + x_4 + x_5 &\leq 35 + 5(1 - y_5) \\
x_2 + x_3 + x_4 + x_5 + x_6 &\leq 35 + 5(1 - y_6) \\
x_3 + x_4 + x_5 + x_6 + x_7 &\leq 35 + 5(1 - y_7) \\
x_4 + x_5 + x_6 + x_7 + x_1 &\leq 35 + 5(1 - y_1) \\
x_5 + x_6 + x_7 + x_1 + x_2 &\leq 35 + 5(1 - y_2) \\
x_6 + x_7 + x_1 + x_2 + x_3 &\leq 35 + 5(1 - y_3) \\
x_7 + x_1 + x_2 + x_3 + x_4 &\leq 35 + 5(1 - y_4) \\
y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 &\geq 3
\end{aligned}
$$

- If the number of workers that start on a Monday exceed 10 then the number of workers available on a Wednesday should be no more than 28:

$$
\begin{aligned}
x_1 &\leq 10 + 30z \\
x_6 + x_7 + x_1 + x_2 + x_3 &\leq 28 + 12(1 - z)
\end{aligned}
$$

- The number of workers that start on a Saturday should be less than the number of workers that start on a Monday or a Tuesday:

$$
\begin{aligned}
x_6 &\leq x_1 + 40w - 1 \\
x_6 &\leq x_2 + 40(1 - w) - 1
\end{aligned}
$$

- A total 40 workers are available:

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 \leq 40$$

Finally entire problem can be formulated as:

$$\begin{array}{rl}
\min & 300x_1 + 320(x_2 + x_7) + 330(x_3 + x_4 + x_5 + x_6) \\
\text{s.t.} & x_1 + x_2 + x_3 + x_4 + x_5 \geq 25 \\
& x_2 + x_3 + x_4 + x_5 + x_6 \geq 25 \\
& x_3 + x_4 + x_5 + x_6 + x_7 \geq 25 \\
& x_4 + x_5 + x_6 + x_7 + x_1 \geq 25 \\
& x_5 + x_6 + x_7 + x_1 + x_2 \geq 25 \\
& x_6 + x_7 + x_1 + x_2 + x_3 \geq 25 \\
& x_7 + x_1 + x_2 + x_3 + x_4 \geq 25 \\
& x_1 + x_2 + x_3 + x_4 + x_5 \leq 35 + 5(1 - y_5) \\
& x_2 + x_3 + x_4 + x_5 + x_6 \leq 35 + 5(1 - y_6) \\
& x_3 + x_4 + x_5 + x_6 + x_7 \leq 35 + 5(1 - y_7) \\
& x_4 + x_5 + x_6 + x_7 + x_1 \leq 35 + 5(1 - y_1) \\
& x_5 + x_6 + x_7 + x_1 + x_2 \leq 35 + 5(1 - y_2) \\
& x_6 + x_7 + x_1 + x_2 + x_3 \leq 35 + 5(1 - y_3) \\
& x_7 + x_1 + x_2 + x_3 + x_4 \leq 35 + 5(1 - y_4) \\
& y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 \geq 3 \\
& x_1 \leq 10 + 30z \\
& x_6 + x_7 + x_1 + x_2 + x_3 \leq 28 + 12(1 - z) \\
& x_6 \leq x_1 + 40w - 1 \\
& x_6 \leq x_2 + 40(1 - w) - 1 \\
& x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 \leq 40 \\
& x_i \in \mathbb{Z}_+, \ \forall i = 1, \ldots, 7 \\
& y_i \in \{0, 1\}, \ \forall i = 1, \ldots, 7 \\
& z, w \in \{0, 1\}
\end{array}$$

2. Jack and Jill love to watch movies and, as a couple, have vowed to always watch their movies together as a form of spending time with each other. They have subscribed for a 1 month free trial and estimated that they will watch 20 movies by the end of the free trial. They have pre-selected 100 movies that they would like to watch and now want to decide which 20 out of these 100 to watch during the free trial month. These 100 movies are divided as follows: Movies $1 - 20$ are action movies, Movies $21 - 40$

are romantic comedies, Movies $41-60$ are TV series, Movies $61-80$ are documentaries and Movies $81-100$ are science fiction.

They associated an estimated common satisfaction index s_i to each movie and wish to maximize their estimated common satisfaction during this month. They also established the following constraints so that each of them will not be too unhappy: They must choose at least 3 action movies and at least 3 romantic comedies, but not more than 5 science fiction and no more than 2 documentaries.

(assume that they will be able to watch all movies they request)

a) Write an Integer programming formulation that Jack and Jill can use to solve this problem of maximizing their estimated common satisfaction. Clearly indicate your decision variables and what they mean.

Solution: Let $x_i \in \{0,1\}$ indicates whether they chose movie i (it is equal to 1 if movie is chosen). Then problem can be formulated as:

$$\begin{aligned}
\min \quad & \sum_{i=1}^{100} x_i s_i \\
\text{s.t.} \quad & \sum_{i=1}^{100} x_i = 20, \\
& \sum_{i=1}^{20} x_i \geq 3 \\
& \sum_{i=21}^{40} x_i \geq 3 \\
& \sum_{i=61}^{80} x_i \leq 2 \\
& \sum_{i=81}^{100} x_i \leq 5 \\
& x_i \in \{0,1\}, \forall i = 1, \ldots 100.
\end{aligned}$$

b) Write one or more linear constraints to express the following additional conditions: (you may add decision variables if necessary, as long as you make it clear what they mean. Also, if you use a Big-M constant, please provide your best estimates of their values.)

i. Since movies 11, 12 and 13 are part of a series, either they get all of them, or they get none of them.

Solution: Constraint is saying that either $x_{11} = x_{12} = x_{13} = 1$ or $x_{11} = x_{12} = x_{13} = 0$. In other words:

$$x_{11} = x_{12} = x_{13}$$

or

$$x11 - x_{12} = 0$$
$$x_{11} - x_{13} = 0.$$

ii. Movies 41–50 are respectively seasons 1–10 of Friends, so they have decided that if they watch one season of Friends, they must watch all seasons before it. For example, if they watch season 6, they must also watch seasons 1–5.
Solution: In other words if season $i \in \{2, 3, \ldots, 10\}$ is watched then seasons $1, 2, \ldots, i-1$ must also be watched. Effectively, for every $i \in \{2, 10\}$ we want to have constraint: if $x_{40+i} = 1$ then $x_{40+j} = 1$ for all $j = 0, 1, \ldots, i-1$. We can model it as:

$$\sum_{j=0}^{i-1} x_{40+1} = (i-1)x_{40+i}, \ \forall i = 2, \ldots, 10$$

For example for $i = 6$ above constraint becomes:

$$x_{40} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} \geq 5x_{46}$$

If they watch season 6, then we have $x_{46} = 1$. Above constraint becomes:

$$x_{40} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} \geq 5$$

thus forcing $x_{40} = x_{41} = x_{42} = x_{43} = x_{44} = x_{45}$. On the other hand if they do not watch season 6, we have $x_{46} = 0$ and constraint becomes trivial.

iii. If they watch at least 2 movies out of 24, 34 and 77, then they must not watch more than one movie out of 27, 35 and 79.
Solution: In other words if $x_{24} + x_{34} + x_{77} \geq 2$ then $x_{27} + x_{35} + x_{79} \leq 1$, which we can model as:

$$x_{24} + x_{34} + x_{77} \leq 2 + y$$
$$x_{27} + x_{35} + x_{79} \leq 1 + 2(1 - y)$$

If they watch 2 or 3 movies out of movies 24, 34 and 77 then first constraint will force $y = 1$, and second constraint will become:

$$x_{27} + x_{35} + x_{79} \leq 1$$

If they watch only one of the movies 24, 34 and 77 or none, then first constraint will be nonbinding for y.

3. A power plant has four boilers. If a given boiler is operated, it can be used to produce a quantity of steam (in tons) between the minimum and maximum given in Table 1. The cost of producing a ton of steam on each boiler is also given, as well as the fixed cost of operating each boiler. Steam from the boilers is used to produce power on three turbines. If operated, each turbine can process an amount of steam (in tons) between the minimum and maximum given in Table 2. The cost of processing a ton of steam and the power produced by each turbine is also given. The power plant must produce at least 9,000 Kwh of power.

Boiler #	Min.	Max.	Cost per ton ($)	Fixed cost ($)
1	400	900	9	160
2	500	700	7	200
3	300	600	8	190
4	240	800	6	250

Table 1: Data for each of the boilers

Turbine #	Min.	Max.	Kwh per ton of steam	Processing Cost per Ton ($)
1	100	700	7	9
2	400	800	3	4
3	300	600	6	6

Table 2: Data for each of the turbines

Formulate a mixed integer linear program that can be used to minimize the cost while satisfying all constraints. Clearly specify what are the decision variables (and what they mean), as well as the objective function and constraints. Also, if you use a Big-M constant, please provide your best estimates of their values.

Solution:

Let's introduce following variables:

- $x_i,\ i = 1, 2, 3, 4$ denote quantity of steam produced by boiler i.
- $y_i \in \{0, 1\},\ i = 1, 2, 3, 4$ indicates whether boiler i is operated.
- $z_i,\ i = 1, 2, 3$ denote quantity of steam processed by turbine i.
- $w_i \in \{0, 1\},\ i = 1, 2, 3$ indicates whether turbine i is operated.

then the problem can be formulated as:

$$
\begin{aligned}
\min \quad & 160y_1 + 200y_2 + 190y_3 + 250y_4 + 9x_1 + 7x_2 + 8x_3 + 6x_4 + 7z_1 + 3z_2 + 6z_3 \\
\text{s.t.} \quad & 400y_1 \leq x_1 \leq 900y_1 \\
& 500y_2 \leq x_2 \leq 700y_2 \\
& 300y_3 \leq x_3 \leq 600y_3 \\
& 240y_4 \leq x_4 \leq 800y_4 \\
& 100w_1 \leq z_1 \leq 700w_1, \\
& 400w_2 \leq z_2 \leq 800w_2, \\
& 300w_3 \leq z_3 \leq 600w_3, \\
& x_1 + x_2 + x_3 + x_4 = y_1 + y_2 + y_3 \\
& 7z_1 + 3z_2 + 6z_3 \geq 9000 \\
& y_i \in \{0,1\}, i = 1,2,3,4 \\
& w_i \in \{0,1\}, i \in 1,2,3
\end{aligned}
$$

4. Consider the polytope P given by:

$$
\begin{aligned}
2x_1 + 3x_2 &\leq 100 \\
x_1 &\geq 0 \\
x_2 &\geq 0.
\end{aligned}
$$

Notice that the point $(5,10)$ is a feasible point for P. Write an integer programming formulation whose feasible set is exactly the set of integer points in P except $(5,10)$. (No need to give any objective function.)

Solution: First note that:

$$
\begin{aligned}
(P \cap \mathbb{Z}) \setminus \{(5,10)\} &= \{(x_1,x_2) \in \mathbb{Z} : 2x_1 + 3x_2 \leq 100, x_1 \geq 0, x_2 \geq 0\} \setminus \{(5,10)\} = \\
&= \{(x_1,x_2) \in \mathbb{Z} : 2x_1 + 3x_2 \leq 100, x_1 \geq 0, x_2 \geq 0\} \cap \\
&\quad (\{(x_1,x_2) : x_1 \leq 4\} \cup \{(x_1,x_2) : x_1 \geq 11\} \cup \\
&\quad \{(x_1,x_2) : x_2 \leq 9\} \cup \{(x_1,x_2) : x_2 \geq 11\})
\end{aligned}
$$

This we can model as:

$$2x_1 + x_2 \leq 100 \tag{1}$$
$$x_1 \geq 0,\ x_2 \geq 0 \tag{2}$$
$$x_1 \leq 4 + M(1 - y_1) \tag{3}$$
$$x_1 \geq 6 - M(1 - y_2) \tag{4}$$
$$x_2 \leq 9 + M(1 - y_3) \tag{5}$$
$$x_2 \geq 11 - M(1 - y_4) \tag{6}$$
$$y_1 + y_2 + y_3 + y_4 \geq 1 \tag{7}$$
$$x_1, x_2 \in \mathbb{Z},\ y_1, y_2, y_3, y_4 \in \{0, 1\} \tag{8}$$

Constraint (7) will force at least one y_i to be 1, which will result in at least one of the constraints (3) - (6) being binding, thus excluding point $(5, 10.)$ Since all other feasible points will satisfy at least one of the constraints (3) - (6), they will be included.